ClusterFusion

project

Presented By:

MD Morshaline Mojumder

BSSE-1504

IIT,Du

Supervisor:

Dr.Sumon Ahmed

Associate Professor

IIT,Du

"Good morning everyone. My name is morshaline, and I am
presenting my project under the guidance of my supervisor,
Dr.sumon ahmed. The topic of my project is Text
Summarization."

Overview

Text Summarization is the process of condensing a large text into a shorter version while retaining its key information. It helps extract the most important points from lengthy documents, making them easier to read and understand .

The tool utilizes the following extractive approaches for text summarization:

Word Frequency Algorithm: This approach ranks sentences based on the frequency of important words in each sentence. The sentences with higher frequencies of essential words are considered more relevant and are included in the summary and it has 5 steps implementation

1. Tokenization: Breaking the document into sentences/words.

2. Create the word frequency table

3. Score the sentences: Term frequency : score a sentence by its words, adding the frequency of every non-stop word in a sentence.

4.Find the threshold : we are considering the average score of the sentences as a threshold.

5.Generate the summary :Select a sentence for a summarization, If the sentence score is more than the average score.

Text Rank Algorithm: Text Rank treats each sentence as a node in a graph. The algorithm evaluates the importance of sentences by considering the relationships between them, and the most significant sentences are selected to form the summary.

Functions related to TextRank:

- **TFIDFCalc**: Calculates the Term Frequency-Inverse Document Frequency (TF-IDF) for each word, helping to understand how important a word is within the context of the document.

- **CosineSimilarity**: Measures the similarity between two sentences based on their word vectors, which is essential for creating the sentence similarity graph in TextRank.

- **scoreSentences()**: Scores each sentence based on how central or important it is within the document, considering the word frequencies and sentence similarities.

- **generateSummary()**: Uses the sentence scores to generate a summary, selecting the most relevant sentences based on the TextRank

Gaussian Mixture Model (GMM) is a **probabilistic clustering model and** is a type of **soft clustering**, meaning a data point can belong to multiple clusters with different probabilities. clustering helps organize documents, sentences, or words into meaningful groups.

Relationship Between Gaussian Mixture Model (GMM), TF-IDF, and Cosine Similarity:

TF-IDF (Term Frequency-Inverse Document Frequency) is a numerical representation of text used to identify important words in a document.

How does GMM use TF-IDF?

1. **Convert text into TF-IDF vectors**, where each document is represented as a vector.
2. **Apply GMM to cluster the documents** based on their vector representations.
3. **Each cluster represents a topic**, helping in text clustering and summarization.

Cosine Similarity measures how similar two text vectors are by calculating the cosine of the angle between them.

How does GMM use Cosine Similarity?

1. **Compute Cosine Similarity between sentences to form a similarity matrix.**
2. **Use this similarity matrix to refine GMM clustering.**
3. **Ensure that similar sentences are placed in the same cluster.**

https://github.com/morshaline1504